

Tensor computations

Subquadratic overparameterization for iterative decomposition of smoothed tensors

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because a uniform convergence guarantee must handle arbitrary smoothed factors, random initialization and near-exact recovery below quadratic overparameterization; broad importance includes nonconvex optimization and machine learning.

Problem statement

Given $n \geq r$, take arbitrary base factors $\bar{A}, \bar{B}, \bar{C} \in \mathbb{R}^{n \times r}$, and form A, B, C by adding independent Gaussian entries of variance ρ^2/n , where $\rho = r^{-q}$ and $q > 0$ is fixed. The input is

$$T = \sum_{j=1}^r a_j \otimes b_j \otimes c_j.$$

Does there exist $k(r) = o(r^2)$, a polynomial lower bound $n_0(r)$, and an explicitly specified ALS or gradient-descent method, initialized randomly, such that for every $n \geq n_0(r)$ and $0 < \varepsilon < 1$ it returns k summands with

$$\left\| T - \sum_{i=1}^k x_i \otimes y_i \otimes z_i \right\|_F \leq \varepsilon \|T\|_F$$

using polynomially many exact real arithmetic operations in $n, r, \log(1/\varepsilon)$, with probability $1 - o_r(1)$ over the once-drawn smoothed input and at least inverse-polynomial conditional success over initialization? Update rules, initialization, and permitted restarts must be specified. Initial factors must be sampled independently of the input and its unknown factors, apart from an explicitly specified scalar scale normalization and dependence on n, r, ρ, ε . The method must optimize the squared residual by alternating block least squares or gradient steps; substituting an algebraic tensor-recovery algorithm does not answer the question. The base-factor quantifier must be preserved rather than replaced by assumptions about the successful trajectory.

References

Dionysis Arvanitakis, Vaidehi Srinivas, and Aravindan Vijayaraghavan, *Open Problem: How Much Overparametrization Is Needed for ALS in Tensor Decomposition?*, COLT 2026, PMLR 336, 7105–7110, §2, Open Problem 2 and the following probability clarification. Zhang et al., *VALG: An Agentic System for ML Theory Research and Demonstrations on COLT 2026 Open Problems*, §4.1.1, Theorem 4.1 and Discussion.

Status check — 2026-09-10

Rechecked the [COLT 2026 problem](#) and [VALG, §4.1.1](#), and searched for subquadratic ALS resolutions. VALG explicitly reports only partial progress, with $k = \Theta(r^{5/3}(\log r)^{5/2})$, adding scale, interference, balance, smoothing and dimension assumptions and using a proposal/landing/ALS pipeline. It is not a verified solution under every initialization and update convention displayed here; arbitrary base factors remain the decisive unresolved quantifier. No full resolution was located. The source informally motivates well-conditioned factors, but its displayed smoothed model permits arbitrary base factors; this entry preserves that quantifier.